ProofOfWork ��� Report

ProofOfWork — Project Report

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1. Overview

ProofOfWork is a personal portfolio delivered as a single, self-contained web application. It presents its author's profile, academics, projects, achievements and certificates through an animated, app-like interface — with no frameworks, no build tooling and no server. The entire experience ships in one `index.html` plus a folder of static assets.

2. Motivation

Most student portfolios are static, templated pages. The goal here was different: build something that itself demonstrates engineering ability — smooth animation, live data, a sensible architecture and clean UX — so the portfolio is, quite literally, proof of work.

3. Features

- Animated splash screen with a self-drawing SVG logo.
- A swipeable, tabbed section carousel (Profile, Academics, Projects, Achievements, Certificates) with scroll-snap navigation.
- Live GitHub integration: a self-updating contribution graph rendered and animated from real data, plus an auto-fetched list of the latest repositories.
- Detailed project pages, each with a description, an animated language-usage graph, technology chips, a screenshot gallery, and links to the live app, source, slides, README and report.
- A built-in "Ask about Manthan" chatbot that answers questions about the author and the projects.
- Light/dark theming, full responsiveness and PWA-readiness.

4. Architecture

The application is intentionally dependency-free. State lives in a single JavaScript data structure describing the profile and each project; the UI is generated from that data at runtime. Screens are plain DOM sections toggled by a lightweight router, and all animation is done with CSS transitions/keyframes plus small amounts of JavaScript. External data (GitHub contributions and repositories) is fetched client-side with graceful fallbacks, so the app degrades cleanly with no network.

5. Technology Stack

Layer	Technology
Markup	HTML5
Styling	CSS3 (custom properties, grid/flex, keyframe animation)
Logic	Vanilla JavaScript (ES6)
Graphics	Inline SVG
Data	GitHub REST API
Type	Google Fonts (Space Grotesk, Inter, JetBrains Mono)
Delivery	Fully static, deployable to any host

6. Implementation Highlights

- **Data-driven UI:** adding a project is as simple as appending one object to a data array.
- **Live contribution graph:** a 53-week grid is built by date and coloured by activity level, then animated in as a wave with a continuous shimmer and a glow on active days.

- **Resilience:** every network feature has a fallback, so nothing breaks offline.
- **Performance:** no framework overhead — the app is a single file that loads instantly.

7. Status & Future Work

The portfolio is complete and actively maintained. Planned additions include shareable deep-links to individual projects and sections, full PWA installability, and an optional LLM-backed version of the chatbot via a lightweight serverless proxy.

8. Conclusion

ProofOfWork packages a person's work into a fast, polished, framework-free application that doubles as a demonstration of front-end engineering — living up to its name.